
Topics In Trustworthy Machine Learning : The Statistical Fairness-Accuracy Frontier, Fallah et al. [2026]

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Introduction

Algorithmic fairness has become a central concern in machine learning, as predictive models increasingly guide high-stakes decisions in lending, hiring, and healthcare. A fundamental challenge arises when a single predictive model must serve multiple demographic groups simultaneously.

In the M2 IASD course on Topics in Trustworthy Machine Learning, we studied this tension through fair classification under demographic parity. Via a Lagrangian relaxation, we derived the optimal fair classifier and statistical guarantees on both fairness and excess risk. The paper by Fallah et al. [2026] tackles an analogous but complementary question in a regression setting. Rather than enforcing a hard fairness constraint, it studies the full range of fairness-accuracy tradeoffs via a Pareto frontier. The main contributions are minimax-optimal estimators, optimal sample allocation strategies, and uniform finite-sample bounds showing that estimation error impacts each group differently.

1 FA frontier and λ -parametrization

Fallah et al. [2026] consider a prediction problem with two groups $g \in \{r, b\}$ and a hypothesis class $\mathcal{H}$. For any predictor $f \in \mathcal{H}$, define the group-wise risks $R_g(f) := \mathbb{E}_g[\ell(f(X), Y)]$.

The planner faces a tradeoff between accuracy across groups: improving performance for one group may degrade it for the other. This tradeoff is captured by the *fairness-accuracy (FA) frontier*, defined by Liang et al. [2024] as the set of Pareto-optimal risk pairs $(R_r(f), R_b(f))$.

A convenient way to parameterize this frontier is via a scalar $\lambda \in [0, 1]$, representing the relative importance assigned to group r . The planner minimizes a weighted risk and selects

$$f_\lambda := \arg \min_{f \in \mathcal{H}} \lambda R_r(f) + (1 - \lambda) R_b(f).$$

As λ varies, the solutions f_λ trace out the FA frontier. The endpoints $\lambda = 1$ and $\lambda = 0$ correspond to the group-specific optimal predictors for groups r and b , respectively, while intermediate values yield compromise solutions.

While the FA frontier provides a clean population-level characterization of the tradeoff, it does not address how the optimal predictor f_λ can be estimated from finite data. In practice, the planner must learn from samples drawn from each group, introducing statistical uncertainty that can distort both accuracy and fairness.

The central question of Fallah et al. [2026] is therefore: *how does estimation affect the fairness-accuracy tradeoff?*

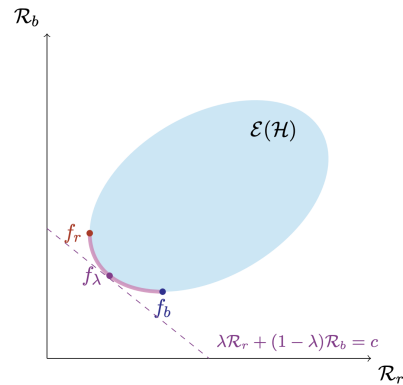


Figure 1: Illustration of FA frontier, parametrized by λ .

(Taken from Fallah et al. [2026])

2 The Linear Model Class: A Deliberate Restriction

The FA frontier developed in section 1 is abstract: it applies to any model class and any loss. To move from qualitative insights to exact estimators, Fallah et al. [2026] restricts attention to linear regression with squared loss. This is not a limitation but a deliberate choice: in this setting the risk decomposes as a Mahalanobis distance (making the geometry of the frontier explicit).

2.1 Model Specification

The prediction class is the set of linear functions $\mathcal{H} = \{f_\beta(x) = x^\top \beta : \beta \in \mathbb{R}^d\}$, and the loss is the squared error $\ell(f_\beta(X), Y) = (\beta^\top X - Y)^2$. Despite the groups having different data-generating processes, the planner is constrained to use a *single* predictor β for everyone, reflecting legal prohibitions on group-specific models.

Formally, the paper considers the model class $\mathcal{P}_{\text{linear}}(\sigma^2)$, consisting of all distributions in which, for each group $g \in \{r, b\}$, the outcome Y given covariate $X \in \mathbb{R}^d$ satisfies:

$$Y = X^\top \beta_g + \varepsilon_g, \quad (1)$$

for some parameter $\beta_g \in \mathbb{R}^d$ and a random variable ε_g such that $\mathbb{E}[\varepsilon_g | X] = 0$ and $\mathbb{E}[\varepsilon_g^2 | X] = \sigma^2$. When the group-conditional covariate distribution P_X^g is known, we write $\mathcal{P}_{\text{linear}}(P_X^r, P_X^b, \sigma^2)$.

Two features of this specification deserve brief comment. First, the regression vector β_g differs across groups, which is what makes the fairness problem non-trivial: if $\beta_r = \beta_b$, no tension arises. Second, the noise is *homoskedastic* (σ^2 is the same for both groups and independent of X), which ensures the problem is *group-balanced*: the fairest predictor lies between the two group optima on the frontier, producing a clean, symmetric tradeoff.

2.2 Population Quantities

Under this model, the group- g population covariance and cross-moment are:

$$\Sigma_g := \mathbb{E}_g[XX^\top], \quad \nu_g := \mathbb{E}_g[XY] = \Sigma_g \beta_g,$$

with empirical counterparts $\hat{\Sigma}_g := \frac{1}{n_g} \sum_{i=1}^{n_g} X_i^{(g)} X_i^{(g)\top}$ and $\hat{\nu}_g := \frac{1}{n_g} \sum_{i=1}^{n_g} Y_i^{(g)} X_i^{(g)}$.

The group- g risk of any linear predictor β takes the simple form: $R_g(\beta) = \|\beta - \beta_g\|_{\Sigma_g}^2 + \sigma^2$, where $\|v\|_{\Sigma_g}^2 := v^\top \Sigma_g v$ is the Mahalanobis norm.

2.3 Closed Form for the Frontier

Recall from section 1 that the planner's target is $\beta_\lambda = \arg \min_\beta R_\lambda(\beta)$. Under the linear model, this admits a closed form:

$$\beta_\lambda = \Sigma_\lambda^{-1} \nu_\lambda, \quad \Sigma_\lambda := \lambda \Sigma_r + (1 - \lambda) \Sigma_b, \quad \nu_\lambda := \lambda \nu_r + (1 - \lambda) \nu_b. \quad (2)$$

Note that β_λ is *not* a simple convex combination $\lambda \beta_r + (1 - \lambda) \beta_b$: the mixing happens at the level of (Σ_g, ν_g) , and the covariance matrices mediate how the two groups' parameters are blended. This is why knowledge of Σ_r, Σ_b matters so much.

2.4 Regularity Assumptions

The theoretical analysis relies on four assumptions, which we briefly discuss.

Assumption 1–2 (Invertible covariances) For each group g , Σ_g is invertible (ensuring β_g is the unique risk minimizer) and $\hat{\Sigma}_g$ is invertible a.s. (which requires $n_g \geq d$, guaranteeing that $\hat{\beta}_g = \hat{\Sigma}_g^{-1} \hat{\nu}_g$ is well-defined).

Assumption 3–4 (Two-sided spectral control) For each group g , the covariates satisfy a small-ball condition $P_g(|\theta^\top X| \leq t \|\theta\|_{\Sigma_g}) \leq (C_g t)^{\alpha_g}$ for all $\theta \neq 0$, $t > 0$, and a sub-Gaussian tail bound $P_g(|X^\top u| \geq t \|u\|_{\Sigma_g}) \leq 2 \exp(-t^2/K_g^2)$ for all $u \in \mathbb{R}^d$. Together, these control the smallest and largest eigenvalues of $\hat{\Sigma}_g$; both hold for Gaussian covariates.

Assumption 5 (Subgaussian noise) For both groups $g \in \{r, b\}$, the noise ε_g is conditionally mean-zero and σ_ε -subgaussian given X , independent of X . This assumption is needed only for the uniform frontier bound (Theorem 2).

3 Estimation with Known Covariances

We begin with the more favorable setting: the planner knows Σ_r and Σ_b (e.g., from large administrative datasets of covariates), but must estimate β_r, β_b from limited labeled data.

3.1 The Optimal Estimator

The proposed estimator in the paper has an elegant structure. It plugs in OLS estimates $\hat{\beta}_g$ for the unknown β_g while retaining the true covariances:

$$\tilde{\beta}_\lambda = \Sigma_\lambda^{-1} (\lambda \Sigma_r \hat{\beta}_r + (1 - \lambda) \Sigma_b \hat{\beta}_b). \quad (3)$$

Using the *true* Σ_λ avoids the bias from covariance misestimation.

3.2 Minimax Optimality

The paper proves that $\tilde{\beta}_\lambda$ is **minimax-optimal**:

Theorem 1 (Minimax Optimality) Under Assumptions 1–2, $\tilde{\beta}_\lambda$ achieves the infimum in the minimax problem

$$\inf_{\tilde{\beta}} \sup_{P \in \mathcal{P}_{\text{linear}}(P_X^r, P_X^b, \sigma^2)} \mathbb{E}[\|\tilde{\beta} - \beta_\lambda\|_{\Sigma_g}^2],$$

simultaneously for each group $g \in \{r, b\}$ and hence also for the weighted excess risk R_λ .

That is, its worst-case error matches the fundamental lower bound that no estimator can beat.

3.3 Simplified Bound and Optimal Sample Allocation

If we limit to spherical covariances $\Sigma_g = \rho_g^2 I_d$, we have the following bound for the excess risk :

$$\mathbb{E}[R_\lambda(\tilde{\beta}_\lambda) - R_\lambda(\beta_\lambda)] \leq \frac{2\sigma^2 d}{\lambda \rho_r^2 + (1 - \lambda) \rho_b^2} \left(\frac{\lambda^2 C'_r \rho_r^2}{n_r} + \frac{(1 - \lambda)^2 C'_b \rho_b^2}{n_b} \right), \quad (4)$$

where C'_g depends on the small-ball parameters. The excess risk grows linearly in d , decreases with n_r, n_b , and each group's contribution is weighted by λ^2 and ρ_g^2 .

For a fixed budget $n_r + n_b$, minimizing (4) via Cauchy–Schwarz yields:

$$\frac{n_r}{n_b} = \frac{\lambda \rho_r}{(1 - \lambda) \rho_b}. \quad (5)$$

This is a *Neyman allocation* adapted to the fairness setting: invest more samples in the group with higher welfare weight λ and more dispersed covariates ρ_g , equalizing the variance-weighted marginal value of information across groups.

Asymmetric Per-Group Impact

Even with the optimal estimator, the group with larger covariate variance ρ_g^2 suffers a strictly larger welfare loss, independently of sample balance.

4 Estimation with Unknown Covariances

We now turn to the more realistic setting where neither β_g nor Σ_g are known. The natural estimator is OLS on the weighted empirical objective:

$$\hat{\beta}_\lambda = \hat{\Sigma}_\lambda^{-1} \hat{\nu}_\lambda = (\lambda \hat{\Sigma}_r + (1 - \lambda) \hat{\Sigma}_b)^{-1} (\lambda \hat{\nu}_r + (1 - \lambda) \hat{\nu}_b). \quad (6)$$

4.1 The Bias–Variance Decomposition

The paper’s central structural result for this setting is:

$$\mathbb{E}[\|\hat{\beta}_\lambda - \beta_\lambda\|_{\Sigma_g}^2] = \underbrace{V_g(\lambda)}_{\text{variance}} + \underbrace{B_g(\lambda)}_{\text{bias}}. \quad (7)$$

The variance term $V_g(\lambda)$ has the same structure as in the known-covariance case (equation 4) : it reflects the irreducible sampling uncertainty about β_r and β_b .

Bias term $B_g(\lambda)$: a genuinely new phenomenon absent in the known-covariance case. It arises because $\beta_\lambda = \Sigma_\lambda^{-1}\nu_\lambda$ is a covariance-weighted blend, so misestimating Σ_g displaces the planner to a *different point on the frontier*—a welfare distortion from informational asymmetry:

$$B_g(\lambda) \asymp \frac{\lambda^2(1-\lambda)^2 \rho_g^2 \rho_r^4 \rho_b^4 d}{(\lambda \rho_r^2 + (1-\lambda) \rho_b^2)^4} \left(\frac{1}{n_r} + \frac{1}{n_b} \right) \|\beta_r - \beta_b\|^2.$$

Three features are noteworthy. First, B_g is proportional to $\|\beta_r - \beta_b\|^2$: the more the groups differ, the costlier the misestimation; when $\beta_r = \beta_b$, the bias vanishes. Second, the factor $\lambda^2(1-\lambda)^2$ makes it vanish at the endpoints $\lambda \in \{0, 1\}$ —at $\lambda = 1$, the estimator reduces to $\hat{\beta}_r$ which does not depend on Σ_b . Third, matching upper and lower bounds (Theorems 2–3 of Fallah et al. [2026]) establish that the OLS estimator is near-minimax-optimal up to constant factors.

Impact on Optimal Sampling

The coexistence of bias and variance creates a tension:

Term	Optimal allocation	Dominates when
Variance V_g	$n_r/n_b = \lambda \rho_r / ((1-\lambda) \rho_b)$	$\ \beta_r - \beta_b\ $ small
Bias B_g	$n_r \approx n_b$ (balanced)	$\ \beta_r - \beta_b\ $ large

When groups are heterogeneous (precisely when fairness is most pressing) bias dominates and the planner should collect data more evenly, even if λ would otherwise favor one group. Beyond sampling design, covariance misestimation introduces a structural distortion in how errors are distributed across groups. Even if β_r and β_b were known, the planner cannot recover $\beta_\lambda = \Sigma_\lambda^{-1}\nu_\lambda$ without knowing Σ_r, Σ_b : the target is a covariance-weighted blend, not a simple average. Replacing the true covariances by their empirical counterparts therefore lands the estimator at a displaced point on the frontier. This displacement is asymmetric: its magnitude scales as $\lambda(1-\lambda)^2$ for group r and $\lambda^2(1-\lambda)$ for group b . As a consequence, the empirical frontier is not merely pushed outward uniformly, but *rotated*: when λ is small, group r bears a disproportionate share of the estimation error, and conversely for group b when λ is large. A planner who sets λ to reflect her fairness preferences may therefore realize, in finite samples, a strictly different welfare allocation than intended.

4.2 Uniform Frontier Bounds

The preceding analysis fixes λ . In practice a designer may wish to inspect the entire empirical frontier before committing to a welfare weight, using the same dataset. Because empirical errors at different λ values are computed from the same data, they are highly correlated; a naive union bound over a grid of λ values is both conservative and restricted to finitely many points.

The paper resolves this by viewing $\Delta_\lambda := \hat{R}_\lambda - R_\lambda \in \mathbb{R}^2$ as a stochastic process indexed by $\lambda \in [0, 1]$, and establishing a *uniform* high-probability bound via a Doob martingale argument combined with Dudley’s chaining inequality.

Theorem 2 (Uniform frontier bound, Theorem 4 of Fallah et al. [2026]) *Under Assumptions 1–5, for any $\alpha > 2$ and $\delta \in (0, 1)$, with probability at least $1 - \delta - \eta$,*

$$\sup_{\lambda \in [0, 1]} \|\Delta_\lambda\|_2 \leq C_u \log\left(\frac{1}{\delta}\right) \left(\frac{\log(n_r)^5}{\sqrt{n_r}} + \frac{\log(n_b)^5}{\sqrt{n_b}} \right),$$

where $\eta \leq (n_r + n_b)^{-C_\alpha} + 2^{-\alpha_r n_r/6} + 2^{-\alpha_b n_b/6}$.

This result provides a *confidence band* around the entire empirical FA frontier, valid simultaneously for all λ . Its width shrinks at rate $1/\sqrt{n_r} + 1/\sqrt{n_b}$ (up to logarithmic factors), quantifying how much data is needed to make reliable welfare comparisons between alternative fairness–accuracy tradeoffs. Crucially, a designer can select λ adaptively after inspecting the empirical frontier and still enjoy the same high-probability guarantee, with no additional statistical cost.

5 Experiments

We reproduce the synthetic experiments from Fallah et al. [2026] to compare the population frontier, the known-covariance estimator, and the unknown-covariance estimator.

Setup. We consider a two-group Gaussian linear model in dimension d , with isotropic covariances $\Sigma_g = \sigma_g^2 I_d$. For each group $g \in \{r, b\}$, $X_g \sim \mathcal{N}(0, \Sigma_g)$, $Y_g = X_g^\top \omega_g + \xi$, where $\xi \sim \mathcal{N}(0, 1)$. The parameter vectors (ω_r, ω_b) are constructed by adding a dense random perturbation with fixed norm, ensuring a constant level of heterogeneity $\|\omega_b - \omega_r\|_2$ across Monte Carlo runs.

Protocol. We evaluate the FA frontier for 50 values of $\lambda \in [0, 1]$ and average results over 100 Monte Carlo repetitions. For each λ , we compute the pair (R_r, R_b) under the true data-generating distribution.

Results. Figure 2 shows that estimation induces a contraction of the FA frontier toward the center. This effect is more pronounced when sample sizes are small or covariances differ across groups. The unknown-covariance estimator exhibits larger deviations from the population frontier, highlighting the additional difficulty of estimating group covariances.

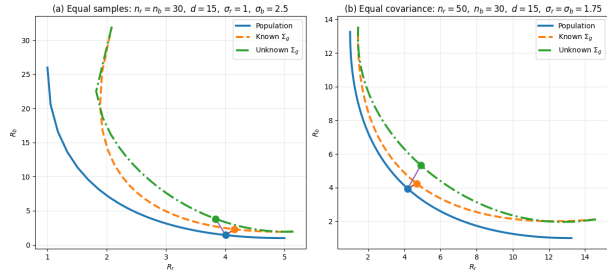


Figure 2: Empirical FA frontiers averaged over 100 Monte Carlo runs. *Left:* equal samples ($n_r = n_b = 30$), unequal covariances ($\rho_b > \rho_r$), $d = 15$. *Right:* unequal samples ($n_r = 50, n_b = 30$), equal covariances, $d = 15$.

6 Conclusion

This work provides a principled statistical perspective on fairness–accuracy tradeoffs in multi-group prediction by characterizing its frontier and studying its estimation in finite samples.

In the linear regression setting, the analysis reveals a structured bias–variance tradeoff, where covariance estimation introduces a bias term that is most pronounced when groups are heterogeneous. This leads to non-trivial implications for data collection, notably the tension between variance-optimal and fairness-aware sampling strategies.

Beyond its specific assumptions, the paper highlights a broader message: fairness is not only a modeling or optimization issue, but also a statistical estimation problem. Understanding how finite-sample effects reshape fairness guarantees is therefore essential for deploying reliable and equitable machine learning systems.

Extending these insights beyond linear models and squared loss remains an important direction for future work.

References

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